Lecture Notes on the IEEE 9-Bus AC Power-Flow Problem

Generated from the loadflow-lite MATPOWER fixtures

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Abstract

The IEEE 9-bus system is a compact benchmark for the steady-state AC power-flow problem. It is small enough to inspect by hand, but it still contains the main modeling features that make power flow nontrivial: multiple generator buses, voltage magnitude control, nonlinear active- and reactive-power balance equations, branch losses, and reactive-power exchange with transmission lines. These notes formulate the problem, describe the specific 9-bus network used in this repository, discuss the solved operating point recorded in the checked-in MATPOWER-derived fixtures, and offer an economic interpretation for readers whose primary background is markets rather than power engineering.

1 Context and Relevance

Power-flow analysis is the basic steady-state calculation behind many power-system engineering tasks. Given a network model and specified generation, load, and voltage-control data, the AC power-flow problem computes the complex voltage at each bus and the real and reactive power flows on each branch. The result is not a dynamic trajectory; it is a snapshot of an operating point under balanced, sinusoidal, three-phase conditions.

This calculation matters because many operational and planning decisions depend on quantities that are not specified directly in the input data: voltage magnitudes at load buses, bus voltage angles, slack-bus real-power output, generator reactive-power output, branch flows, and system losses. A candidate dispatch may satisfy a real-power energy balance in aggregate while still producing unacceptable voltage profiles, excessive line flows, or reactive-power requirements. AC power flow is therefore the entry point for contingency analysis, voltage-security assessment, optimal power flow, state estimation validation, and many simulation studies.

It is also the feasibility check that sits underneath every market-clearing calculation in wholesale electricity markets. Locational marginal prices (LMPs) are the dual variables on the nodal energy-balance constraints in optimal power flow; congestion rents are the dual variables on the binding line constraints; the slack-bus convention plays the role of a numéraire. Most production-grade market software runs a linearized *DC* power-flow approximation of the equations developed here, but the AC problem is the ground truth those approximations are calibrated against. Understanding the AC equations on a small, transparent case is therefore useful background for anyone working with electricity-market models.

The case used here is the WSCC 3-machine 9-bus system, often called the Anderson 9-bus after its appearance in Anderson and Fouad's *Power System Control and Stability*. It is useful pedagogically because it is small but not degenerate. It has three generator buses (one slack, two PV), six PQ buses, and nine branches; impedances are reported on a 100 MVA system base. The topology is a meshed transmission network rather than a radial feeder, and the solution exhibits both real-power losses and significant reactive-power effects from line charging.

2 A Glossary for Readers from Economics

This note assumes familiarity with linear algebra and Newton's method but not with power engineering. The vocabulary is the main barrier. The following one-line glosses cover the terms that appear repeatedly. An economic analogue is given where one is genuinely close.

Bus. A node in the network — physically, a substation busbar. Read as "node" throughout.

Branch. An edge between two buses. May be a transmission line or a transformer. Read as "arc" or "link."

Per-unit (p.u.). A non-dimensionalization. Every quantity is divided by a reference base value, so a voltage of 1.00 p.u. means "the nominal design voltage at this bus," and a power of 1.00 p.u. means 100 MW on the system used here (100 MVA base). Treat it exactly like reporting macro variables as fractions of GDP — the base is sometimes implicit but always there.

Real power (MW). The energy actually delivered per unit time; what consumers pay for and what generators are dispatched on.

Reactive power (MVAr). A network-support quantity needed to sustain AC voltages in the presence of inductive and capacitive elements. It does not deliver useful work in the same sense as real power, but transmission cannot operate without it. In markets it is procured as a separate ancillary service. Treat as an unpriced-or-separately-priced complement to real power.

Slack bus. The bus where voltage magnitude and angle are fixed by convention and the generator output is whatever the rest of the problem implies. Plays exactly the role of a numéraire in general equilibrium: one node's price is fixed at 1 (here, one angle is fixed at 0); all other angles are then *relative* to it.

PV bus. A generator bus with specified real-power output and voltage magnitude (the "P" and the "V"). Reactive output and angle are solved. Read as "a node that targets a price-like variable (voltage), at whatever quantity-support is required."

PQ bus. A bus with specified real and reactive demand (the "P" and the "Q"). Both voltage magnitude and angle are solved. Read as "a passive consumption node."

Voltage angle (θ_i). A scalar at each node. Differences in θ across a branch drive real-power flow on that branch. The cleanest economic analogy is to relative prices: angle differences are to AC power flow what price differences are to commodity arbitrage.

Series reactance (x) and series resistance (r). Per-branch impedance parameters. r produces real-power losses (the "transaction cost" interpretation); x governs how strongly an angle difference pushes real power across the branch.

Shunt charging (b). A line-side capacitive element that injects reactive power into the network. It is a "free" source of reactive support, in contrast to generator-supplied reactive power which has positive marginal cost.

Tap ratio, phase shift. Transformer settings. Both equal 1 and 0 respectively in this fixture, so they can be ignored here.

Generator step-up transformer (GSU). The transformer that connects a generator to the high-voltage transmission grid. The IEEE 9-bus has three of them (branches 1–4, 3–6, 8–2).

A useful one-liner: *angles are like prices, real power is the commodity, reactive power is the lubricant, losses are the transaction cost, and the slack bus is the numéraire.*

3 Mathematical Formulation

3.1 Network Model

Let the system have n buses and complex bus-voltage phasors

$$V_i = |V_i| e^{j\theta_i}, \qquad i = 1, \dots, n.$$

The nodal admittance matrix is

$$Y_{\text{bus}} = G + jB,$$

where G_{ik} and B_{ik} are the conductance and susceptance entries of the bus admittance matrix — effectively the entries of the network's weighted Laplacian, with weights set by line impedances. At each bus, the net complex power injection is

$$S_i = P_i + jQ_i = V_i I_i^*,$$

where P_i is real power and Q_i is reactive power. Using $I = Y_{\text{bus}} V$, the standard polar-form AC power-flow equations are

$$P_i = |V_i| \sum_{k=1}^{n} |V_k| \left[G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k) \right], \tag{1}$$

$$Q_i = |V_i| \sum_{k=1}^{n} |V_k| \left[G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k) \right]. \tag{2}$$

The specified net injections are formed from positive generation and positive load:

$$P_i^{\text{spec}} = P_{G,i} - P_{D,i}, \qquad Q_i^{\text{spec}} = Q_{G,i} - Q_{D,i}.$$

All real and reactive powers in these notes are reported on the 100 MVA base used by the fixture files unless explicitly shown in MW or MVAr.

3.2 Bus Types and Unknowns

The power-flow equations are solved with different specifications at different bus types:

- **Slack (reference) bus**: specifies $|V|$ and θ. Its real and reactive generation are outputs of the solution. The slack bus serves two structural roles. First, it provides the angle reference $\theta_1 = 0$, without which the system is invariant under a global phase shift and the equations are underdetermined — the numéraire role. Second, it closes the real- and reactive-power balances: total losses (real and reactive) depend on the solved state and cannot be assigned to a generator in advance, so the residual is absorbed at a single chosen bus.

- **PV bus**: specifies real-power generation P_G and voltage magnitude $|V|$. The voltage angle θ and the reactive generation Q_G are both solved. It is therefore wrong to read the case-side Q_G at a PV bus as a setpoint; in the fixtures it is just an initial value that the solver overwrites.

- **PQ bus**: specifies real and reactive load. Both $|V|$ and θ are solved.

The resulting nonlinear system is square: there are $n-1$ active-power equations (one per non-slack bus) and $n - n_{\mathrm{PV}} - 1$ reactive-power equations (one per PQ bus). It is solved by Newton's method on the mismatch residual; the standard derivation, including the four Jacobian sub-blocks, is sketched in Appendix A.

4 The Specific IEEE 9-Bus Network

The fixture used here is `data/case9.json`, with solved values from `data/case9_solution.json`. The case has three generator buses: bus 1 is the slack bus, while buses 2 and 3 are PV buses. Loads are located at buses 5, 7, and 9. The total load is

$$P_D = 315.00 \text{ MW}, \qquad Q_D = 115.00 \text{ MVAr}.$$

The case-side P_G at the slack bus and Q_G at all generator buses are MATPOWER initial values rather than scheduled targets, so it is the solved generator outputs that should be compared against the load. Only P_G at the two PV buses (163.00 and 85.00 MW) is a true setpoint, and the solver honors both exactly.

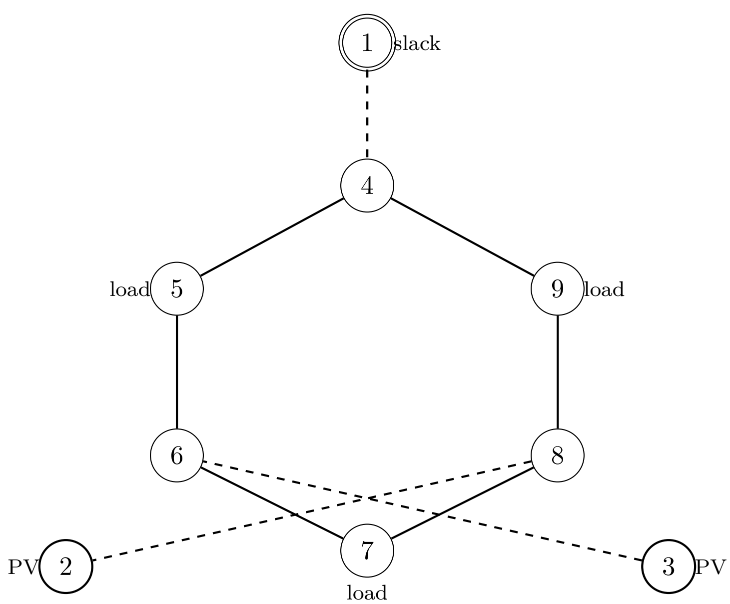

Figure 1: Topology of the IEEE 9-bus (WSCC) system. The double-circle node is the slack bus; thick-circle nodes are PV generator buses; "load" annotations mark the three PQ buses with non-zero demand. Dashed lines (1–4, 3–6, 8–2) are the generator step-up transformer branches; solid lines are transmission branches. The interior loop 4−5−6−7−8−9−4 is the meshed sub-network through which generator power redistributes.

Topologically, the meshed sub-network is the loop

$$4 \to 5 \to 6 \to 7 \to 8 \to 9 \to 4.$$

Each generator feeds into this loop through a single radial branch: gen 1 via branch 1–4, gen 2 via branch 8–2, and gen 3 via branch 3–6. On the radial hops there is no choice of path, but once

Table 1: Bus data and solved voltages for the IEEE 9-bus case.

Bus	Type	P_D MW	Q_D MVAr	P_G^{case} MW	Q_G^{case} MVAr	$\lvert V\rvert$ p.u.	θ deg	P_G^{sol} MW
1	slack	0.00	0.00	72.30	27.03	1.0400	0.0000	71.64
2	PV	0.00	0.00	163.00	6.54	1.0250	9.2800	163.00
3	PV	0.00	0.00	85.00	-10.95	1.0250	4.6648	85.00
4	PQ	0.00	0.00	0.00	0.00	1.0258	-2.2168	0.00
5	PQ	90.00	30.00	0.00	0.00	1.0127	-3.6874	0.00
6	PQ	0.00	0.00	0.00	0.00	1.0324	1.9667	0.00
7	PQ	100.00	35.00	0.00	0.00	1.0159	0.7275	0.00
8	PQ	0.00	0.00	0.00	0.00	1.0258	3.7197	0.00
9	PQ	125.00	50.00	0.00	0.00	0.9956	-3.9888	0.00

Table 2: Branch parameters for the 9-bus case. All series and charging parameters are in p.u. on the system base; `tap_ratio` = 1.0 and `phase_shift` = 0 for every branch in this fixture, so those columns are omitted. Branches with $r = 0$ and $b = 0$ (1–4, 3–6, 8–2) are the generator step-up transformers in the original WSCC model; the fixture stores them as plain reactances because per-unit base voltages are not tracked here.

Branch	From	To	r	x	b
1	1	4	0.00000	0.05760	0.00000
2	4	5	0.01700	0.09200	0.15800
3	5	6	0.03900	0.17000	0.35800
4	3	6	0.00000	0.05860	0.00000
5	6	7	0.01190	0.10080	0.20900
6	7	8	0.00850	0.07200	0.14900
7	8	2	0.00000	0.06250	0.00000
8	8	9	0.03200	0.16100	0.30600
9	9	4	0.01000	0.08500	0.17600

power enters the loop it divides according to the network impedances and the bus voltage-angle profile. This is the structural reason the angle profile matters: it is the state that determines how much real power takes each direction around the loop.

5 Discussion of the Solved Operating Point

5.1 Active-power balance and losses

The solved generator outputs are

$$P_G^{\text{sol}} = 319.641 \text{ MW},$$
$$Q_G^{\text{sol}} = 22.840 \text{ MVAr}.$$

Since the total real load is 315.000 MW, the real-power loss is

$$P_{\text{loss}} = 319.641 - 315.000 = 4.641 \text{ MW},$$

which equals the sum of $P_f + P_t$ over all branches as it must. The slack bus supplies 71.641 MW; the PV buses hold their scheduled real-power outputs of 163.000 MW at bus 2 and 85.000 MW at bus 3.

5.2 Voltage profile

The voltage magnitudes remain close to nominal. The highest solved voltage is the slack-bus voltage, 1.0400 p.u.. The lowest voltage is at bus 9, 0.9956 p.u., which is also the largest load bus (125 MW and 50 MVAr). The drop is driven primarily by the local reactive demand: the sensitivity $\partial|V|/\partial Q$ is strongly negative at heavily loaded buses, and bus 9 has both the largest Q_D and only two relatively long branches connecting it to the rest of the network.

A subtler observation is that bus 6 sits at 1.0324 p.u., which is *above* the PV setpoint of 1.0250 p.u. at the adjacent generator buses 2 and 3. In a radial network this would be unusual, but in a meshed network with significant line charging the shunt b on branches 5–6 and 6–7 injects reactive power into bus 6 and lifts its voltage above the nominal generator setpoint. This is why the PV-setpoint values cannot be read as upper bounds on the system voltage profile.

5.3 Angles and active-power flow

The voltage angles span from $-3.9888°$ at bus 9 to $9.2800°$ at bus 2. In an AC transmission network, active-power transfer is strongly tied to angle differences across each branch through the near-lossless approximation

$$P_{ij} \;\approx\; \frac{|V_i|\,|V_j|}{x_{ij}}\,\sin(\theta_i - \theta_j).$$

Branch 8–2 illustrates this clearly. The fixture stores the orientation as 8 to 2, with $P_f = -163.000$ MW and $P_t = 163.000$ MW, so the real-power flow is from bus 2 toward bus 8. Plugging in $|V_2| = 1.0250$, $|V_8| = 1.0258$, $x_{8\text{-}2} = 0.0625$, and $\theta_2 - \theta_8 = 9.2800° - 3.7197° = 5.5603°$:

$$P_{28} \;\approx\; \frac{1.0250 \times 1.0258}{0.0625}\,\sin(5.5603°) \;=\; 1.6303 \text{ p.u.} \;=\; 163.03 \text{ MW},$$

which matches the solved 163.00 MW to three figures. The lossless approximation is essentially exact on this branch because $r = 0$. Reading the table this way — always check angle differences when interpreting branch flows — is the most useful intuition the 9-bus case offers.

5.4 Reactive-power balance

Table 3: Solved branch powers. The "from" and "to" quantities follow the fixture branch orientation; negative values mean flow enters that end of the branch.

From	To	P_f MW	Q_f MVAr	P_t MW	Q_t MVAr	P_ℓ MW	Q_ℓ MVAr
1	4	71.641	27.046	-71.641	-23.923	0.000	3.123
4	5	30.704	1.030	-30.537	-16.543	0.166	-15.513
5	6	-59.463	-13.457	60.817	-18.075	1.354	-31.531
3	6	85.000	-10.860	-85.000	14.955	0.000	4.096
6	7	24.183	3.120	-24.095	-24.296	0.088	-21.176
7	8	-75.905	-10.704	76.380	-0.797	0.475	-11.502
8	2	-163.000	9.178	163.000	6.654	0.000	15.832
8	9	86.620	-8.381	-84.320	-11.313	2.300	-19.694
9	4	-40.680	-38.687	40.937	22.893	0.258	-15.794

The sign discussion generalizes: on branch 7–8, the negative P_f indicates that real power enters the branch at bus 8 and leaves at bus 7, even though the fixture orientation runs the other way.

The reactive-power column tells a different story. Summing $Q_f + Q_t$ over all branches gives exactly

$$\sum_{\text{branches}} (Q_f + Q_t) \;=\; Q_G^{\text{sol}} - Q_D \;=\; 22.840 - 115.000 \;=\; -92.160 \text{ MVAr}.$$

This is not a coincidence and not a violation of conservation; it is the reactive analogue of the active-loss identity, but with a sign that can go either way. Several branches inject more reactive power through their shunt charging elements than they absorb in their series reactances, so the per-branch "Q loss" can be negative. In this case the line charging dominates: the network as a whole is a net reactive-power *source*, supplying 92.160 MVAr that the generators do not have to provide. This is one of the reasons AC power flow must model reactive power explicitly rather than treating it as a small correction to a real-power-only calculation.

5.5 Economic interpretation

For readers approaching this from electricity markets rather than power engineering, four observations on the same numbers are worth pulling out explicitly.

Losses are a wedge embedded in prices. The 4.641 MW of real losses is generation that has to be procured but that no consumer pays for at a meter. In wholesale markets, this is socialized through the loss component of the locational marginal price. In nodal LMP decompositions, $\text{LMP}_i = \lambda_{\text{energy}} + \text{loss}_i + \text{congestion}_i$; the 4.641 MW corresponds to the system-wide loss component of that decomposition. Losses are a real economic cost; they just don't appear on the demand side of the energy-balance constraint.

Reactive power is largely an unpriced ancillary service. The 92.160 MVAr that the network supplies from line charging is, in effect, free reactive support: there is no marginal-cost generator behind it. The 22.840 MVAr from generators is a separate matter — it occupies generator capacity that would otherwise be available for real power and is typically procured through reactive-support markets or VAR-obligation tariffs. The sharp split between "free reactive from shunt charging" and "costly reactive from generators" is the microeconomic content of the reactive-power balance.

Angle differences are shadow prices in linearized form. The DC power-flow approximation, which underlies most production-grade market software, replaces the AC equations with a linearization in which $P_{ij} \propto (\theta_i - \theta_j)/x_{ij}$. In that linearization the angles *are* the dual variables of the nodal balance constraints — up to the constant of proportionality, they are prices. Worked example: branch 8–2 carries 163 MW at an angle wedge of 5.56°, exactly tracking the lossless formula. In the DC-OPF analogue this would translate to a price wedge between buses 2 and 8. The structural insight from the AC calculation — that real power is driven by angle differences — is the same insight that says LMPs in adjacent buses can differ even when no line is at its capacity limit, because losses depend on flow.

Bus 9 is electrically remote, and that has a price story. Bus 9 has the lowest voltage (0.9956 p.u.) and the most negative angle (−3.99°), and it is at the corner of the loop opposite both the slack bus and most of the generation. In the LMP picture, electrical remoteness translates directly to price separation: the same physical factors that drive bus 9's voltage down and its angle most negative would, in a market dispatch, give it a higher LMP than the rest of the network whenever load near bus 9 grows or the connecting branches become congested. "Distance from supply" is a power-flow concept and a price concept simultaneously.

6 Takeaways

1. The IEEE 9-bus problem is a minimal but meaningful AC power-flow benchmark: it includes slack, PV, and PQ bus behavior in a small meshed network with three generator step-up transformers and a six-bus transmission loop.

2. The unknown state consists of voltage angles at non-slack buses and voltage magnitudes at PQ buses. Slack-bus real and reactive power and PV-bus reactive power are all solved outputs, not specified inputs; case-side P_G at the slack and Q_G at every generator should be read as initial values, not setpoints.

3. The solved real-power generation exceeds total real load by 4.641 MW, which is the network real-power loss at this operating point.

4. Voltage angle differences govern active-power flows through the near-lossless relation $P_{ij} \approx (|V_i||V_j|/x_{ij})\sin(\theta_i - \theta_j)$. Branch 8–2 reproduces its solved 163 MW flow from this formula to three figures.

5. Branch-flow signs must be read relative to the stored branch orientation. A negative "from" power means the actual real-power flow is opposite to the stored from-to direction.

6. Reactive-power behavior is qualitatively different from real-power loss accounting because line charging can inject reactive power. In this case the network is a net reactive-power source of 92.160 MVAr, and an interior PQ bus (bus 6) sits above the adjacent PV setpoint as a result.

7. The vocabulary maps cleanly onto market concepts: angles $\leftrightarrow$ prices, real power $\leftrightarrow$ commodity, reactive power $\leftrightarrow$ ancillary service, losses $\leftrightarrow$ transaction cost, slack bus $\leftrightarrow$ numéraire. AC power flow is the feasibility layer beneath every LMP calculation.

A Newton–Raphson Solution

The non-slack bus active-power mismatches and the PQ-bus reactive-power mismatches are

$$\Delta P_i = P_i^{\text{spec}} - P_i(V, \theta), \quad i \notin \{\text{slack}\},$$

$$\Delta Q_i = Q_i^{\text{spec}} - Q_i(V, \theta), \quad i \in \{\text{PQ}\}.$$

With n_{PV} PV buses, the unknowns are the $n-1$ non-slack voltage angles and the $n - n_{\text{PV}} - 1$ PQ-bus voltage magnitudes, giving a square nonlinear system. Newton–Raphson linearizes the residual using the Jacobian

$$J = \begin{bmatrix} \frac{\partial P}{\partial \theta} & \frac{\partial P}{\partial |V|} \\ \frac{\partial Q}{\partial \theta} & \frac{\partial Q}{\partial |V|} \end{bmatrix} = \begin{bmatrix} H & N \\ M & L \end{bmatrix},$$

restricted to the rows and columns corresponding to the unknowns above, and updates

$$\begin{bmatrix} \Delta\theta \\ \Delta|V| \end{bmatrix} = J^{-1} \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}.$$

Iteration continues until $\max_i(|\Delta P_i|, |\Delta Q_i|)$ falls below a tolerance, typically 10^{-6} to 10^{-8} p.u. The MATPOWER fixture used here does not record the iteration count or final mismatch, so we cite only the converged state.